3. An individual withdrawing the maximum sustainable constant inflation-adjusted amount from a portfolio at the start of each year over a thirty-year retirement period.

Exhibit 3.9 provides the distribution of results for the simulations. These simulations are based on a standard 50/50 portfolio using historical Morningstar data for the S&P 500 and intermediate-term government bonds. For the lump-sum investment, the numbers represent the distribution of average compounded returns over one hundred thousand thirty-year periods. For the accumulation phase, the distribution of outcomes is for the internal rate of return for the final wealth accumulation when making thirty annual contributions. For the retirement phase, the distribution of results is for the internal rates of return on the portfolio when withdrawing the maximum sustainable amount over a thirty-year period with distributions taken at the start of each year.

Exhibit 3.9

Distribution of Compounded Real Returns over 30 Years

Monte Carlo Simulations for a 50/50 Asset Allocation

Based on SBBI Data, 1926–2016, S&P 500 and Intermediate-Term Government Bonds